

1. Administrative & Study Identification

Full Study Title: A Phase 1 Open-label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of MK-4700 as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced or Metastatic Solid Tumors
Short Title/Acronym: MK-4700-001
Protocol Number: 4700-001
NCT Number: NCT06894771
Sponsor Name: Merck / Merck Sharp & Dohme LLC
Investigational Product: MK-4700 and Pembrolizumab (MK-3475 / KEYTRUDA®)
Phase of Development: Phase 1
Trial Status: Active, not recruiting (ACTIVE_NOT_RECRUITING)
Medical Monitor: Medical Director, Merck Sharp & Dohme LLC
Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of different escalating doses of MK-4700 given alone and in combination with pembrolizumab, and to identify a safe and tolerated dose that may work to treat certain types of cancer.
Secondary Objectives: To measure the pharmacokinetic profile of MK-4700 and to determine what happens in a person's body over time when the drug is administered.

2.2 Background & Rationale To address the clinical gap for patients with advanced or metastatic solid tumors who have experienced disease progression on or after prior anti-cancer treatments, or who have been intolerant to/refused all treatments known to confer clinical benefit. The study aims to determine if the novel targeted approach of MK-4700, alone or combined with the immunotherapy pembrolizumab, enhances the immune system's ability to combat specific cancers.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-group/Parallel-group (Arm 1: MK-4700 alone; Arm 2: MK-4700 + Pembrolizumab) **Blinding:** Open-label **Randomization:** N/A (Dose Escalation)

3.2 Planned Enrollment & Duration Target \$\$\$: 82 participants
Number of Sites: Multicenter (e.g., Hackensack University Medical

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has histologically or cytologically confirmed advanced/metastatic solid tumor by pathology report who have experienced disease progression on or after prior anti-cancer treatments, or been intolerant to, or refused all treatment known to confer clinical benefit. 2. Has head and neck squamous cell carcinoma (HNSCC), melanoma (cutaneous), non-small cell lung cancer (NSCLC), cervical cancer, triple negative breast cancer (TNBC), urothelial carcinoma (UC), or renal cell carcinoma (RCC; clear cell, papillary). 3. If human immunodeficiency virus (HIV) infected, must have well controlled HIV on antiretroviral therapy (ART). 4. Has normal cardiac function.

4.2 Exclusion Criteria 1. If HIV-infected with a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 2. Has known additional malignancy that is progressing or has required active treatment within the past 2 years. 3. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 4. Has active autoimmune disease that has required systemic treatment in the past 2 years except replacement therapy. 5. Has history of (noninfectious) pneumonitis/interstitial lung disease that required steroids. 6. Has current pneumonitis/interstitial lung disease. 7. Has active infection requiring systemic therapy. 8. Has known history of Hepatitis B (defined as Hepatitis B surface antigen reactive) or known active Hepatitis C virus infection. 9. Has had an allogeneic tissue/solid organ transplant in the last 5 years.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive escalating doses of MK-4700 every three weeks (Q3W). Participants in Arm 2 receive MK-4700 in combination with pembrolizumab Q3W. Eligible participants in Arm 1 who experience progressive disease (PD) may cross over to Arm 2. **Route of Administration:** Intravenous / Oral (per trial protocol administration guidelines for MK-4700 and pembrolizumab). **Duration of Treatment:** Maximum of 35 cycles (approximately 2 years; cycles are 21 days in length) or until PD, death, toxicity, or withdrawal of consent. Total study duration including follow up is up to ~4.5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Antiretroviral therapy (ART) for well-controlled HIV; replacement therapies for specific autoimmune conditions. **Prohibited:** Systemic steroid treatments for pneumonitis; systemic treatment for active

autoimmune disease (within past 2 years); systemic therapy for active infections.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	2-4 weeks prior	Informed Consent, Medical History, Tumor Pathology Confirmation, Cardiac Evaluation
Baseline	Day 0	First Dose (Cycle 1), Baseline PK sampling
Interim	Every 21 Days	Q3W Treatment Administration, Dose-Limiting Toxicity (DLT) Monitoring, PK/PD Blood Sampling, AE Assessment
End of Study	Follow-up	Final Safety Evaluations, PD Assessment, Survival Follow-up (Up to ~4.5 years)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition:

- Dose-Limiting Toxicity (DLT):** The occurrence of any of the following toxicities during Cycle 1, considered related to study intervention administration:
 - Grade 4 nonhematologic toxicity
 - Grade 4 hematologic toxicity lasting ≥ 7 days, except thrombocytopenia
 - Any nonhematologic AE $\geq$ Grade 3 in severity, with exceptions
 - Any Grade 3 or Grade 4 nonhematologic laboratory value, with pre-specified exceptions
 - Any Grade 3 or Grade 4 laboratory abnormalities, with exceptions
 - Febrile neutropenia Grade 3 or Grade 4
 - Prolonged delay (>2 weeks) in initiating Cycle 2 due to intervention-related toxicity
 - Any study drug toxicity that causes the participant to discontinue study drug during Cycle 1
 - Missing $>25\%$ of MK-4700 doses as a result of drug-related AEs during the first cycle
 - Grade 5 toxicity
- Percentage of Participants Who Experience an Adverse Event (AE):** Any untoward medical occurrence temporally associated with the use of study intervention.
- Percentage of Participants who Discontinue Study Treatment Due to an AE.**

Measurement Method: Clinical observation and laboratory evaluations assessed by Common Terminology Criteria for Adverse Events (CTCAE).

7.2 Secondary & Exploratory Endpoints

- Area Under the Concentration-Time Curve (AUC) of MK-4700.
- Maximum Plasma Concentration (C_{max}) of MK-4700.
- Minimum Plasma Concentration (C_{min}) of MK-4700.
- Pharmacodynamics and overall therapeutic response.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of any untoward medical occurrence in a clinical study participant, temporally associated

with the use of study intervention. **Serious Adverse Events (SAEs):** Continuous monitoring for Grade 3/Grade 4 laboratory abnormalities, prolonged delay (>2 weeks) in initiating Cycle 2, and any Grade 5 toxicities. **Data Monitoring Committee (DMC):** Internal safety review committee to evaluate DLTs prior to each dose-escalation phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all participants who received at least one dose of the investigational product).

Sample Size Justification: A Phase 1 dose-escalation design appropriately powered to determine the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D). **Handling of Missing Data:** Participants missing $>25\%$ of MK-4700 doses in the first cycle as a result of drug-related AEs are evaluated specifically for DLT thresholds.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring of safety data, DLT occurrences, and adherence to dose-escalation rules.

Audit Trail: Standard tracking of eCRF, protocol deviations, and centralized laboratory assessments.

11. Study Identifiers & Governance

Primary ID: MK-4700-001 **Registry Number:** NCT06894771 **Sponsor/Lead Organization:** Merck / Merck Sharp & Dohme LLC **Study Title:** A Study of MK-4700 Alone or With Pembrolizumab in Participants With Advanced or Metastatic Solid Tumors (MK-4700-001) **Official Regulatory Title:** A Phase 1 Open-label, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of MK-4700 as Monotherapy and in Combination With Pembrolizumab in Participants With Advanced or Metastatic Solid Tumors

12. Clinical Framework (Oncology Specific)

Primary Condition: Malignant Neoplasms / Advanced or Metastatic Solid Tumors (including HNSCC, Melanoma, NSCLC, Cervical Cancer, TNBC, UC, RCC) **Diagnosis Threshold:** Histologically or cytologically confirmed disease progression on/after prior treatments, or intolerance/refusal of standard treatments **Medical Methodology:** Targeted therapy + Immunotherapy combination

Optimization Target: Identification of Maximum Tolerated Dose (MTD) / Recommended Phase 2 Dose

13. Study Procedures & Methodology

Management Pathway: Dose-escalation rules with a crossover allowed from monotherapy (Arm 1) to combination therapy (Arm 2) upon progressive disease
Education Protocol (HCP): Protocol training on DLT definitions and AE management
Education Protocol (Patient): Informed consent regarding frequent Q3W clinic visits, PK blood draws, and reporting side effects
Quality Audit Mechanism: Continuous safety data reviews prior to proceeding to subsequent dose cohorts

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 4.5 years
Onsite Visit Cadence: Every 3 weeks (Q3W) for up to 35 cycles (approx. 2 years)
Remote Monitoring Frequency: Between treatment cycles to evaluate ongoing AEs
Checklist Review Interval: Pre-dose safety and laboratory reviews prior to every 21-day cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				